ROYAL CANADIAN NAVY

QUALIFICATION STANDARD AND PLAN

Maritime Surface and Sub-surface Officer (MARS) (00207, 00207-13 and CIC 00232-01)

ORCA OIC Class Package (AJLR)

MITE Course IDs:

ORCA OIC Class Package - 117306

ORCA OIC Board and Performance Check - 117503

RECORD OF AMENDMENTS

The “Authority” reference indicates approval of the corresponding chapters of this QSP document. Authority for Chapters 1 and 2 of this QSP resides with Chief of Maritime Staff (CMS) DMTE. The Curriculum Control Authority (CCA) for Chapters 3 and 4 is NOTC. The approval date, the date that the document is checked back in, should also appear in the bottom right corner of every page (the change being made in the footer). The Revision column is intended to provide a short indication of where the change(s) was/were made. Details on the change(s), if applicable, must be amplified in Annex J “Course History”.

REVISION	APPROVAL DATE	AUTHORITY
1.0	12 Sep 2007	CMS/DMTE
2.0	23 Apr 2008	CMS/DMTE
2.1	06 Mar 2015	CNPTG
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FOREWORD

1. This training document is developed under the authority of the Chief Maritime Staff and in accordance with the following references:

   a. The A-P9-050 Series, Canadian Forces Manual of Individual Training and Education;

   b. A-PD-055-022 / PP-001, Canadian Forces Officer Specifications, Integrated Occupation Specification (IOS); and

   c. OSS (Draft) for ORCA Officer in Charge (OIC).

2. This publication is effective upon receipt. Performance Objectives (POs) are listed in the Table of Contents and are explained fully in Chapter 2. The Lesson Objectives, which specify the standard that trainees must achieve to become qualified, are listed in Chapter 4.

3. Suggestions for amendments must be forwarded through the normal chain of command. Specifically, training establishments are to summarize all requested amendments in writing to the applicable approving authority(ies). Final approval for amendments are as follows:

   a. Chapters 1 and 2 – Chief of Maritime Staff, Director of Maritime Training and Education (DMTE). Suggested amendments must first have the approval of the Curriculum Control Authority Training Establishment (NOTC); and

   b. Chapters 3 and 4 – Commanding Officer, VENTURE, the Naval Officers Training Centre. Training establishments that conduct training and are not the CCA must forward suggested amendments to NOTC for approval.

4. This QSP shall be used as the primary authority governing design, conduct and evaluation of the ORCA OIC Class Package course.

5. The designated CCA(s) for this training is indicated below:

CCA	PHASE	QUALIFICATION
NOTC VENTURE	ORCA OIC Class Package	N/A
CMOG 4	ORCA OIC Board and Performance Check	AJLR

CHAPTER 1: TRAINING MANAGEMENT DETAILS

AUTHORIZATION FOR THIS CHAPTER AND AMENDMENTS MUST BE GRANTED BY COMMANDER NAVAL PERSONNEL AND TRAINING GROUP

AIM

6. The aim of this training is to provide individuals with the knowledge and skills required to perform the duties of an Officer-in-Charge of an ORCA class vessel. Attainment of the ORCA OIC Class Package is achieved through a phased approach, as noted below.

7. The qualification of Officer-in-Charge of an ORCA class vessel consists of:

   a. ORCA Class Package (MITE Code 117306 will be awarded on completion of both Part I and II)

      (1) Part I – Academic (Note);

      (2) Part II – OJTPR;

   b. An Officer-in-Charge Certification Board IAW MARCORD 9-50, Annex E (National Qualification Code 117503 will be awarded on completion of the Certification Board and the Performance Check

Note: The option exists to conduct Part I above by self-study and challenge the Academic EC prior to conducting the OJTPR portion of the Class Package.

8. The specific requirements, by target population, to obtain the qualification of ORCA OIC are iaw the ORCA Class OIC OSS and MARCORD 9-50, Annex E.

COURSE DURATION CALCULATION

9. Course length calculation (IAW MARCORD 9-30):

ESTIMATED TIME REQUIREMENT DURING THE COURSE	
Actual training days (Total)	10 days
Classroom / Simulator	5 days
Afloat/OJTPR Phase	5 days
TAT	0
Travel Time	0
Course Duration (in weeks)	**2 weeks**

RESOURCES

10. The following resources are required:

Dedicated Staff (Ashore)

REQUIRED COURSE STAFFING				
TITLE	**RANK**	**QUALIFICATION**	**QTY**	**REMARKS**
CTO	Lt(N)-LCdr	ORCA OIC	1	

Dedicated Staff (OJTPR Sea Phase)

REQUIRED COURSE STAFFING				
TITLE	RANK	QUALIFICATION	QTY	REMARKS
CTO	Lt(N)-LCdr	ORCA OIC	2	Only 1 req if crse load = 5 or less
ORCA CREW	As per MARCORD 4-26	Only NOTC or CMOG 4 OIC Qualified staff may sign the OJTPR	2	Only 1 req if crse load = 5 or less

Critical Training Resources

CRITICAL TRAINING RESOURCES	NO. OF DAYS
Classroom with chart tables, publications, navigation instruments, IT resources	5
Fully crewed ORCA Class vessel as per MARCORD 4-26. Two ships are required for the Sea Phase at maximum load.	5

COURSE CAPACITY

11. Planning for the conduct of training is based on the following guidelines:

a. Maximum – 10 (see note 1);

b. Optimum – 5; and

c. Minimum – 1.

12. The timings in Chapter 4 are based on the assumption that the course will be loaded with the maximum number of trainees (8), for which two simulator bridges will be available (four trainees per simulator).

COURSE PREREQUISITES

13. This course is available to personnel of the Regular Force, Primary Reserve Force and CIC as follows:

STATUS	PREREQUISITES
Regular Force	BWK
Primary Reserve Force	BWK

CIC, Bos'n QL6A/Reserve QL3 and Clearance Diver QL6B	(TOIC) Certification (Note 1).

Notes: These particular target audiences who have already completed the Tender Charge Course are exempt from the Tender OIC Course but still must complete the ORCA Class package. Additional personnel requiring the qualification in order to perform their duties may be considered for training on a case-by-case basis, as adjudicated by the MA in consultation with the CCA and Operational Authority.

RELATED DOCUMENTS

14. This QSP must be used in conjunction with the Occupation Specialty Specification (OSS) related to ORCA OIC and related Occupation Specifications for the applicable MOSIDs. Other related documents are MARCORD 9-50, Annex E and MARCORD 4-26.

EVALUATION

15. Individual evaluation will be in accordance with the requirements of Chapters 3 and 4 of the QSP.

VALIDATION

16. The training will be validated every five years in accordance with Naval Training Instructions, based on Annex B.

QUALIFICATION

17. MITE code 117306 has been created to manage this course. MARCORD 9-50, Annex E further describes the process to achieve the ORCA OIC certification.

PRIOR LEARNING ASSESSMENT

18. Personnel may apply to the Managing Authority (MA) for a prior learning assessment in order to be excluded from any part of the training. Assessment of such personnel must be done on a case-by-case basis in accordance with A-P9-050-000/PT-012 – Canadian Forces Military Equivalencies Program and MARCORD 9-47.

SPECIAL INSTRUCTIONS

Nil

CHAPTER 2: PERFORMANCE OBJECTIVES

AUTHORIZATION FOR THIS CHAPTER AND AMENDMENTS MUST BE GRANTED BY COMMANDER NAVAL PERSONNEL AND TRAINING GROUP

PO 001 Command ORCA Class Vessel

1. Performance: Command an ORCA Class Vessel as an OIC.

2. Conditions.

   a. Given:

      (1) An ORCA Class vessel with minimum crew (iaw CONOPS); and

      (2) References;

   b. Denied: assistance.

   c. Environmental: At sea or alongside in any environmental conditions within the limitations and restrictions contained in MARCORD 4-26.

3. Standard. IAW applicable references and with due regard to safety, the OIC shall Command an ORCA Class vessel by:

   a. Manoeuvring, to include:

      (1) Handling characteristics;

      (2) Berthing / Slipping Procedures;

      (3) Anchoring;

      (4) Towing;

      (5) Heavy weather considerations; and

      (6) IRB Operations.

   b. Navigating, to include:

      (1) Passage Planning;

      (2) Visual and Radar fixing techniques;

      (3) ARPA / ECPINS operations;

      (4) Tidal calculations; and

      (5) Meteorological data interpretation.

   c. Applying the regulations for the prevention of collision at sea, to include:

      (1) Radar Track plotting;

      (2) Relative Velocity;

(3) AIS;

(4) VHF Radio telecommunications; and

(5) Recognizing the importance of MCTS data.

d. Using combat information equipment, to include:

(1) Radar;

(2) Radio Telecommunication; and

(3) Aids to Navigation.

e. Operating the Bridge Control Console (BCC), to include:

(1) Flash up and shut down;

(2) Manipulating the multiple functions; and

(3) Identifying and responding to engineering emergencies.

f. Directing responses to emergencies, to include:

(1) Fires, floods, casualty and stability crises;

(2) MOB;

(3) Steering Gear Breakdown;

(4) Bomb Threat;

(5) Verification Muster;

(6) Survival Stations;

(7) Environmental incidents;

(8) Manipulating the fire detection and alarm system;

(9) Activating fitted systems in the engine room; and

(10) Isolating power.

g. Exercising command and control, to include:

(1) OIC responsibilities; and

(2) Vessel safety and security;

h. Directing seamanship evolutions, to include:

(1) IRB Operations;

(2) Towing;

(3) Slipping and berthing; and

(4) Anchoring;

4. References: All references are listed in Annex B

5. Specification Task Numbers: All tasks elements listed in Annex C

6. Supporting Skills. N/A

7. Supporting Knowledge. All knowledge elements listed in Annex B

8. Training Limitations. None identified.

CHAPTER 3: ASSESSMENT OF TRAINEE

AUTHORIZATION FOR THIS CHAPTER AND AMENDMENTS MUST BE GRANTED BY THE COMMANDING OFFICER, VENTURE, THE NAVAL OFFICERS TRAINING CENTRE

GENERAL

1. The MITE Code 117306 will be awarded upon successful completion of Phase I – Academic and Phase II – OJTPR. This is an “attend only” course; grades and ranking will not be assigned; however, successful completion of the Enabling Check at the end of Phase I – Academic is a pre-requisite for the Phase II – OJTPR, hence a pre-requisite for the successful completion of the course.

2. Completion of the Class specific OJTPR training package is a pre-requisite to the successful completion of the course. Management of the OJTPR package will be the responsibility of the CCA.

3. Verification of satisfactory completion of the EC and OJTPR will be the responsibility of the operational authority CMOG4 in conjunction with the CCA.

4. The qualifications resulting from completion of training requirements and successful ORCA OIC Certification will be entered into the appropriate qualification tracking mechanism by the CCA.

5. Any amendments to the Enabling Check and OJTPR will be the responsibility of the CCA.

PERFORMANCE CHECKS

6. The ORCA OIC Certification process described in MARCORD 9-50 will be the basis for candidate practical assessment and qualification for the ORCA OIC Class Package course. A candidate must attain a pass during the certification process to be awarded the ORCA OIC Certification MITE Code 47503.

7. Any amendments to the Performance Check for ORCA OIC Certification will be the responsibility of the Operational Authority, CMOG4, in concert with the CCA to ensure consistent alignment with the training standard.

8. Guidance for the preparation of candidate for the ORCA OIC Certification is provided in MARCORD 4-26.

PROGRESS MONITORING

9. Candidate progress shall be continually monitored and recorded to:

    a. provide feedback on effectiveness of training;

    b. provide early warning of candidate difficulty;

    c. provide information to the TRB (if required); and

    d. provide information to assist in the completion of the certificates.

10. The instructor should maintain on file results of:

    a. any significant incident related to a candidate's attitude or professionalism; and

b. any counselling sessions or interviews with the candidate.

11. Quizzes and/or assignments are designed to provide feedback to the instructor and candidate, and shall be used to prepare the candidate for the ORCA OIC OJTPR.

12. Progress monitoring will provide candidates with an assessment of their progress throughout the course and assistance shall be provided to any candidate who experiences difficulty. A candidate who experiences difficulty must be given every change to improve; however, he/she must be informed of the consequences of failure, including disposition options.

UNSATISFACTORY TRAINING PROGRESS

13. Unsatisfactory training performance is indicated by:

a. Failure to complete Phase I – Academic;

b. Failure to complete Phase II – Practical OJTPR;

c. Lack of professional/leadership qualities; or

d. Security and/or safety violation.

TRAINING REVIEW BOARD

14. Training Review Board requirements are contained in MARCORD 9-60 Volume 1. TRB shall be convened for unsatisfactory training performance.

COURSE REPORTS

15. This course being attend only, there is no requisite for Course reports (CF 377), however Course Certificates will be given as per NTI 01/08.

16. Performance on this course will be rated as “Completed.”

EVALUATION MATRIX

PO/EO	PURPOSE	WHERE	WHEN	HOW	RESOURCE REQUIREMENTS	LIMITATIONS
001	Evaluate the candidate's ability to command an ORCA Tender	At sea and alongside	During the ORCA OIC Certification.	Oral Board and Practical Check IAW MARCORD 9-50.	ORCA with minimum manning. OIC Board as per MARCORD 9-50	
001.01	Evaluate the candidate's ability to identify and locate ship equipment and systems, including their characteristics and limitations	Alongside	During the ORCA OIC Course Phase I – Academic or through Self Study Program.	Practical IAW checklist #1. Theory Exam and Practical for those completing the EO through self-study.	ORCA 1 CTO	
001.02	Evaluate the candidate's ability to operate ship equipment and systems	Alongside	During the ORCA OIC Course Phase I – Academic or through Self Study Program.	Practical IAW checklist #1. Theory Exam and Practical for those completing the EO through self-study.	ORCA 1 CTO	
001.03	Evaluate the candidate's ability to apply ship operational safety considerations and procedures.	Alongside and at sea	During the ORCA OIC Course Phase I – Academic or through Self Study Program.	Practical IAW checklist #1. Theory Exam and Practical for those completing the EO through self-study.	ORCA 1 CTO	

PO/EO	PURPOSE	WHERE	WHEN	HOW	RESOURCE REQUIREMENTS	LIMITATIONS
001.04	Evaluate the candidate's ability to conduct ship evolutions.	At sea	During the ORCA OIC Course Phase II – Practical or OJT.	Completion of OJTPR Annex E, and IAW checklists in Annex K.	One ORCA/5 candidates with minimum manning 1 CTO	Need ORCA Class vessel

CHAPTER 4: LESSON SPECIFICATIONS

AUTHORIZATION FOR THIS CHAPTER AND AMENDMENTS MUST BE GRANTED BY THE COMMANDING OFFICER, VENTURE, THE NAVAL OFFICERS TRAINING CENTRE

PO 001 COMMAND ORCA CLASS VESSEL

	EO NUMBER	PERFORMANCE
Phase I – Academic / Self Study Program	001.01	Identify Ship Equipment and Systems
	001.02	Operate Ship Equipment and Systems
	001.03	Apply Ship Operational Safety Consideration and Procedures
Phase II – Practical / OJT	001.04	Conduct Ship Evolutions

EO 001.01 Identify Ship Equipment and Systems

1. Performance: The candidate shall be able to identify ship equipment and systems.

2. Conditions:

   a. Given: An ORCA Class vessel;

   b. Denied:

      (1) Assistance; and

      (2) Ship layout diagrams.

   c. Environmental: n/a

3. Standard: The candidate shall be able to identify and locate ship equipment and systems including:

   a. Bridge / Sensor Systems and layout;

   b. Mast/antenna equipment and layout;

   c. Internal communications systems;

   d. Lights and shapes;

   e. Deck/Hull Systems;

   f. Fitted and portable damage control and firefighting equipment and systems; and

   g. Main propulsion and platform control systems and equipment.

4. Time.

   a. 2 X 50 min Lecture

   b. 1 X 50 min Tour of Vessel

5. Method / Media.

   a. Self study package / Distributed Learning

   b. Demo / practical familiarization

6. References.

   a. C1

7. Training Aids.

a. Know Your Ship Book

b. Power Point Lectures

c. ORCA Class Vessel

8. Test Details.

a. There will be no formal EC for this EO. Formative evaluation will be conducted in the course of the Self-study DL package and required successful completion to progress through the entire e-Learning package. Familiarity with ship systems will be evaluated during ORCA OIC Certification.

9. Remarks.

a. N/A

10. Teaching Points.

a. Bridge / Sensor Systems and layout, including:

(1) Gyro/gyro repeats;

(2) Radars;

(3) Electronic fixing aids;

(4) Automatic identification system (AIS);

(5) Echo sounder;

(6) Speed log;

(7) ECDIS;

(8) Helmsman's console components;

(9) Radios (VHF/HF);

(10) ECPIN;

(11) IPCS components; and

(12) Autopilot;

b. Mast/antenna layout, including:

(1) Radars;

(2) Radio antennas; and

(3) GPS antennas;

c. Internal communications systems, including:

(1) Telephone system;

(2) Main broadcast system;

(3) General alarm system;

(4) Sound/self-powered system;

(5) Sound signalling appliances;

(6) Talk back/broadcast intercom system; and

(7) Loud hailers;

d. Lights and shapes, including:

(1) Navigation lights;

(2) Searchlights;

(3) Signal lights;

(4) Emergency lighting; and

(5) Upper deck lighting.

e. Deck/Hull Systems

(1) Lockers:

(a) Pyro;

(b) Cable;

(c) Cleaning gear; and

(d) HAZMAT.

(2) Watertight integrity features;

(3) Smoke zones and escape routes;

(4) Messing/berthing spaces/heads;

(5) Galley/classroom;

(6) Storerooms and stowage;

(7) Engineering spaces;

(8) Cable locker;

(9) HVAC space;

(10) Anchor windlass/cable arrangement;

(11) Deck fittings, including:

(a) Bollards/hard points;

(b) Fair leads;

(c) Gangway/brow/lines;

(d) Deck equipment storage;

(12) Upper deck connections, including:

(a) Fresh water;

(b) Fuel;

(c) Black and grey water;

(d) Power; and

(e) Telecom.

(13) Gas racks;

(14) Life rafts/life saving equipment;

(15) Cranes and lifting appliances; and

(16) Towing gear and arrangements.

f. Fitted and portable damage control and firefighting equipment and systems, including:

(1) Firemain system, including:

(a) Firefighting pumps;

(b) Isolation valves and hydrants;

(2) Firefighting kit:

(a) First aid extinguishers;

(b) Emergency breathing devices; and

(c) Ancillary equipment.

(3) Fitted flood and spray system;

(4) Automated fire suppression CO_2 systems:

(a) Galley fire suppression system; and

(b) Engineering compartments fire suppression system.

(5) Fire/flood detection sensors;

(6) Bilge and ballast system, including:

(a) Hull and fire pump;

(b) Local and remote activation points;

(c) Bilge suction/discharge/collection;

(d) Portable pumps;

(e) Portable flood control equipment;

(7) Engineering Systems:

(a) Tanks; and

(b) Void spaces;

(8) power generation and distribution system, including:

(a) Generators/alternators; and

(b) UPS;

g. Switchboards/breakers/local controls, including:

(1) Power isolation.

(2) Cathodic protection;

(3) Low voltage systems;

(4) Emergency power;

(5) Hotel services;

(6) Shore systems; and

(7) Batteries and charging systems

h. Auxiliary systems, including:

(1) Fresh water system;

(2) Black water system;

(3) Grey water system;

(4) Garbage system;

(5) HVAC system;

(6) Fuel/lube oil system;

(7) Exhausts; and

(8) Compressed air system; and

i. Steering system, including:

(1) Primary;

(2) Secondary; and

(3) Emergency.

j. Main propulsion and platform control systems and equipment, including:

(1) Engines, including:

(a) Gear box and shafting;

(b) Propellers;

(c) Lube oil and cooling system; and

(d) Generators;

(2) Control systems in local and remote modes, including:

(a) Integrated platform control console (IPCC); and

(b) Bridge control console; and

(3) Machinery alarms and sensors.

EO 001.02 Operate Ship Equipment and Systems

1. Performance: Operate Ship Equipment and Systems.

2. Conditions:

   a. Given:

      (1) An ORCA Class Ship with minimum manning (iaw CONOPS);

      (2) Bridge Pack; and

      (3) References;

   b. Denied: assistance.

   c. Environmental: N/A

3. Standard: IAW applicable references, Bridge Pack and established operational guidelines, the candidate shall be able to operate ship equipment and systems including:

   a. Bridge equipment and systems;

   b. Internal Communication systems;

   c. Damage control and firefighting equipment and systems;

   d. Deck/Hull Systems; and

   e. Engines;

4. Time.

   a. 14 X 50 min lecture

5. Method / Media.

   a. Classroom instruction; and

   b. Hands-on practical.

6. References.

   a. C1,C2,C3,

7. Training Aids.

   a. ORCA;

   b. Powerpoints;

   c. Handbook; and

d. Operational manuals.

8. Test Details.

a. There will be no formal EC for this EO. Familiarity with ship systems operation will be evaluated during ORCA OIC Certification.

9. Remarks.

a. N/A

10. Teaching Points.

a. Bridge equipment and systems, including:

(1) Radar systems;

(2) Radio systems;

(a) VHF;

(b) HF; and

(c) VHF Directional Finder;

(3) ECPIN;

(4) GPS;

(5) Autopilot;

(6) AIS;

(7) IPCS;

(8) Echosounder;

(9) Speed Log;

(10) Meteorological System;

(11) Light panel;

(12) Gyro repeats;

(13) Magnetic compass;

(14) Ship whistle;

b. Internal Communication systems including:

(1) Telephone system;

(2) Main broadcast system;

(3) General alarm system;

(4) Sound/self-powered system;

(5) Emergency Sound signalling appliances; and

(6) Loud hailers.

c. Damage control and firefighting equipment and systems including:

(1) Firemain system;

(a) Fitted flood and spray system;

(b) Automated fire suppression systems:

(2) Fire/flood detection sensors;

(3) Bilge and ballast system, including:

(a) Hull and fire pump;

(b) Portable pumps;

(4) Portable flood control equipment.

d. Deck/Hull Systems including;

(1) Anchor windlass/cable arrangement;

(2) Liferafts/life saving equipment; and

(3) Cranes and lifting appliances.

e. Engines, Generators and Switchboard, including:

(1) Operating parameters; and

(2) Normal and emergency starting/shut down procedures.

11. Training Limitations.

a. Heavy weather shiphandling may only be instructed theoretically, unless weather is permitting and within the limitations and restrictions contained in MARCORD 4-26.

EO 001.03 Apply Ship Operational Safety Consideration and Procedures

1. Performance: Apply Ship Operational Safety Consideration and Procedures

2. Conditions:

   a. Given:

      (1) An ORCA Class Ship with minimum manning (iaw CONOPS);

      (2) Bridge Pack; and

      (3) References;

   b. Denied: assistance.

   c. Environmental: At sea or alongside in any or all environmental conditions within the limitations and restrictions contained in MARCORD 4-26.

3. Standard: IAW applicable references, Bridge Pack and established operational guidelines, the candidate shall be able to apply ship operational safety consideration and procedures including:

   a. Pre-sail / pre-arrival checks;

   b. Standard Operating Procedures;

   c. HAZMAT;

   d. RADAZ; and

   e. Stability considerations;

4. Time.

   a. 9 X 50 min lecture

5. Method / Media.

   a. Classroom instruction; and

   b. Hands-on practical.

6. References.

   a. A1,A2,A3,A4,A5,A6,A7,A8,A9,A10,A11,C1,C2

7. Training Aids.

   a. ORCA;

b. Powerpoints;

c. Handbook; and

d. Operational manuals.

8. Test Details.

a. There will be no formal EC for this EO. Familiarity with ship systems operation will be evaluated during ORCA OIC Certification.

9. Remarks.

a. N/A

10. Teaching Points.

a. Pre-sail / pre-arrival checks:

(1) Pre-sail checks including:

(a) Weather forecast;

(b) Supplies;

(c) Engineering systems readiness;

(d) Night Orders;

(e) Navigation plan / tides and currents;

(f) SSD warning & closing up pipes;

(g) Seamanship readiness;

(h) Personnel readiness;

(i) Traffic;

(j) Bridge equipment calibration and readiness; and

(k) Communication readiness; and

(2) Pre-arrival checks including;

(a) Weather updates (prevailing winds on approach);

(b) Navigation plan / tides and currents;

(c) Traffic;

(d) Berth availability and readiness;

(e) Seamanship readiness;

(f) Engineering systems (emergency) readiness;

(g) Personnel readiness; and

(h) SSD warning & closing up pipes; and

b. Standard Operating Procedures, including:

(1) Personnel Safety;

(2) Accident management and reporting;

(3) Night Order / Night Rounds Book;

(4) Life saving and emergency equipment;

(5) Boat control and safety;

(6) Guardrail, ladders and escape hatches;

(7) Fire precaution and prevention including:

(a) Electrical equipment and applicances;

(b) BBQ equipment; and

(c) Cooking equipment and safety measures;

(8) Fire suppression systems;

(9) Care of SCBA equipment;

(10) Electrical Safety;

(11) Crane Handling Safety;

(12) Confined Space safety;

(13) General security;

(14) Ship evolutions including:

(a) Weather restrictions;

(b) Berthing restrictions and considerations;

(c) Anchoring restrictions and considerations;

(d) Shallow water conditions;

(e) High speed maneouvering and wake restrictions;

(f) Upperdeck personnel;

(g) Boat launch and recovery;

(15) Ship administrative routine including:

(a) Establishment and maintenance of Watch Rotation;

(b) Establishment and maintenance of Watch & Station Bill;

(c) Maintenance of OOW notebook;

(d) Maintenance of Ship's Log;

(e) Maintenance of OIC Night Order book;

(f) Establishment and maintenance of Duty Watches;

(g) Maintenance of Night Rounds book;

(h) Maintenance of NTS Cell phone log;

(i) Completion of Engineering and Seamanship Pre-sail checklists;

(j) Completion of NTS Pre-sail checklist;

(k) Completion of daily Sit-Rep calls to NTS and MOC; and

(l) Maintenance of routine pipes, to include;

(a) Closing up/securing SSD, CP and PSH as applicable;

(b) Warning pipes;

(c) Emergency pipes; and

(d) Informative pipes.

c. HAZMAT including:

(1) Pyrotechnics safety;

(2) Storage, handling and use of flammable liquids; and

(3) Waste disposal;

d. RADHAZ including;

(1) Man aloft;

e. Stability considerations, including:

(1) Effect of various load conditions on trim and manoeuvring; and

(2) Ballasting for heavy weather, high speed maneouvering and icing conditions.

EO 001.04 Conduct Ship Evolutions

1. Performance: Conduct Ship Evolutions

2. Conditions:

   a. Given:

      (1) An ORCA Class Ship with minimum manning (iaw CONOPS);

      (2) Bridge Pack; and

      (3) References;

   b. Denied: assistance.

   c. Environmental: At sea or alongside in any or all environmental conditions within the limitations and restrictions contained in MARCORD 4-26.

3. Standard: IAW applicable references, Bridge Pack and established operational guidelines, the candidate shall be able to conduct ship evolutions including:

   a. Seamanship drills;

   b. Engineering drills;

   c. Damage control drills; and

   d. Environmental spill handling

4. Time.

   a. Developmental: 14 X 50 min practical periods.

   b. Evaluation: 21 X 50 min practical periods.

5. Method / Media.

   a. Classroom instruction; and

   b. Hands-on practical.

6. References.

   a. A1,A2,A3,A4,A5,A6,A7,A8,A9,A10,A11,C1

7. Training Aids.

   a. ORCA;

   b. Powerpoints;

c. Handbook; and

d. Operational manuals;

e. Policy, guidance and SOP manuals.

8. Test Details.

a. There will be no formal EC for this EO. Familiarity with ship evolutions will be evaluated during ORCA OIC Certification.

9. Remarks.

a. N/A

10. Teaching Points.

a. Seamanship drills including:

(1) Launching and recovering RHIB;

(2) Anchoring, mooring and coming to a buoy;

(3) Berthing and Departing;

(4) Man Overboard;

(5) Towing;

(6) Blind pilotage; and

(7) SAR;

b. Engineering drills including:

(1) Steering gear breakdown;

(2) Engineering emergencies, alarms and difficulties;

c. Damage control drills including:

(1) Fire; and

(2) Flood;

d. Environmental spill handling;

PERFORMANCE CHECK – PO 001 ORCA OIC Certification

PART 1 – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

ORCA Class vessel

Arrangements and Procedures

Reserve the ORCA Class vessel through MOG 4

PART II – INSTRUCTIONS GIVEN TO TRAINEE

The Performance Check for PO 001 will be conducted by MOG 4. This will be a practical check on the ORCA Class vessel.

PART III – PERFORMANCE CHECK NOTES

N/A

PART IV – PERFORMANCE CHECKLIST

As per MOG4 assessment checklist.

ANNEX A – MAIN REFERENCES

A1 = CF or other official Canadian government publication

A2 = Second and subsequent references in any category

B = Official Allied Force or Allied government publication

C = Commercial publication

D = Other publications

REF CODE	PUBLICATION	QTY
A1	MOG4 SOPs	
A2	SSOs	
A3	MARCORD 4-26 - Tenders	
A4	MARCORD 9-50 – Sea Command and Charge Qualifications	
A5	MARCORD 42-01 – Navigation and Pilotage	
A6	MARCORD 43-02 – RADHAZ Safety and Policy	
A7	MARCORD 66-05 – HAZMAT Management	
A8	Canadian Navigation Manual	
A9	CFCD 105	
A10	C-23-045-000/AG-001 – Damage Control (Sea) Vol I – Shipboard DC	
A11	C-23-045-000/AG-003 – Damage Control (Sea) Vol II – Ship Stability	
C1	ORCA Vessel Class Manual	
C2	Auto Pilot	
C3	ORCA Class Manual	

ANNEX B – TASK LIST

ITEM	TASK / TASK ELEMENT	DISPOSITION TRG/NO TRG	EO NO. PO 001
AT001	Manoeuvre an ORCA Class Vessel	Trg	EO 001.04
BT002	Navigate an ORCA Class Vessel	No Trg	
CT003	Apply the regulations for the prevention of collision at sea	No Trg	
DT004	Use combat information equipment	Trg	EO 001.02
ET005	Operate Bridge Control Console	Trg	EO 001.02
FT006	Direct Response to Emergencies	Trg	EO 001.04
GT007	Exercise command and control	No Trg	
HT008	Conduct Seamanship Evolutions	Trg	EO 001.04
AK001	Relative velocity	No Trg	
AK002	ORCA Handling Characteristics	Trg	EO 001.04
AK003	Conning, steering and engine orders	Trg	EO 001.04
AK004	Berthing and slipping procedures	Trg	EO 001.04
AK005	Ship handling evolutions	Trg	EO 001.04
AK006	Lines and fenders	Trg	EO 001.01
AK007	Speed/time/distance problems	No Trg	
AK008	Propulsion systems	Trg	EO 001.01
BK009	Navigation passage planning	No Trg	
BK010	Navigation instruments/equipment	Trg	EO 001.01
BK011	Visual fixing techniques	No Trg	
BK012	Radar fixing techniques	Trg	EO 001.04
BK013	ARPA (Automatic Radar Plotting Aid) and contact information	Trg	EO 001.04
BK014	Buoyage system	No Trg	

ITEM	TASK / TASK ELEMENT	DISPOSITION TRG/NO TRG	EO NO. PO 001
BK015	Navigation lights and shapes	No Trg	
BK016	Navigation techniques	No Trg	
BK017	Tidal Calculations	No Trg	
BK018	Blind pilotage organization procedures	Trg	EO 001.03
BK019	MCTS (Maritime Communication and Traffic System) procedures	Trg	EO 001.03
BK020	Meteorological Data Interpretation	No Trg	
BK021	Low visibility precautions	Trg	EO 001.03
BK022	Chart symbology	No Trg	
BK023	Chart work	No Trg	
CK024	Anti-collision manoeuvres	No Trg	
CK025	Regulations preventing collision at sea	No Trg	
DK026	Vessel distress equipment	Trg	EO 001.02
DK027	Radio Telecommunications procedures	No Trg	
DK028	Signal flags and pennants	No Trg	
DK029	References, SOPs and orders relevant to the operation of an ORCA Class vessel	Trg	EO 001.03
DK030	Emergency communications GMDSS (Global Maritime Distress and Safety System)	Trg	EO 001.03
EK031	BCC Capabilities and Limitations	Trg	EO 001.01
EK032	BCC Emergency Response Procedures	Trg	EO 001.01
EK033	BCC Operating Procedures	Trg	EO 001.01
EK034	Ship’s Systems	Trg	EO 001.02
EK035	References, SOPs and orders relevant to the operation of an ORCA Class vessel	Trg	EO 001.03

ITEM	TASK / TASK ELEMENT	DISPOSITION TRG/NO TRG	EO NO. PO 001
FK036	Machinery Emergency Operating Procedures	Trg	EO 001.04
FK037	Fitted and portable firefighting equipment	Trg	EO 001.02
FK038	DC Procedures	Trg	EO 001.04
FK039	Damage control equipment	Trg	EO 001.02
FK040	Life saving equipment	Trg	EO 001.04
FK041	Vessel Stability	Trg	EO 001.03
FK042	Damage control conditions	Trg	EO 001.02
FK043	Command Considerations for Equipment Failures or Limitations	Trg	EO 001.02
FK044	Procedures for Emergency Steering	Trg	EO 001.02
FK045	Significant incident report procedures	Trg	EO 001.03
FK046	Equipment casualty reporting and emergency logistics procedures	Trg	EO 001.03
FK047	Bomb threat procedures	Trg	EO 001.04
FK048	Command considerations related to search and rescue (SAR)	Trg	EO 001.04
FK049	Ship's systems	Trg	EO 001.02
FK050	Ship's layout and compartments	Trg	EO 001.01
FK051	Confined Space Entry Safety Considerations	Trg	EO 001.04
FK052	RADHAZ Considerations	Trg	EO 001.03
FK053	Verification Muster Procedures	Trg	EO 001.04
GK054	Vessel safety and security	Trg	EO 001.03
GK055	Officer in Charge responsibilities	Trg	EO 001.04
GK056	Charge, command and control	Trg	EO 001.04

ITEM	TASK / TASK ELEMENT	DISPOSITION TRG/NO TRG	EO NO. PO 001
GK057	Ship's log and officer of the watch book	Trg	EO 001.03
GK058	Ship's routine	Trg	EO 001.03
GK059	Pre Sail Checklist	Trg	EO 001.03
GK060	Safety and Environmental Management System	Trg	EO 001.04
HK061	IRB operations	Trg	EO 001.04
HK062	Towing Procedures	Trg	EO 001.04
HK063	Mooring and anchoring	Trg	EO 001.04

ANNEX C – QUALIFICATION STANDARD AND PLAN WRITING BOARD RECORD OF PROCEEDINGS

Dates: June 04-15 2007.

Location: NOTC VENTURE

Board Members:

SURNAME	INITIALS	RANK	MOSID	UNIT
Bursey (Chair)	C.	LCdr	00207	DMTE 2
Thomson (OSS SME)	S.G.	LCdr	00207	D Mar Pers
Boisjoli (COO SME)	A.J.F.	Cdr	00207	DMPOR
Ford (OIC SME)	W.R.	Lt(N)	CIC	NOTC
Chura (OIC SME)	J.L.	Lt(N)	00207	NOTC
Folk (Stnds Rep)	R.W.	Lt(N)	00211	NOTC
Taylor	S	Lt	00211	CFFSE
Ewing	D.M.	CPO1	00123	D MAR Pers
McLeod	J.D.R.	CPO2	00123	CFFSE Standards
Pelletier	M	CPO2	00123	MARPAC F44
Tremblay	E.M.	PO2	00122	MOG 4
Ryan	K.G.	MS	00122	MOG 4

Monday June 4 2007

1. Capt(N) Truelove provided opening remarks to help focus the ORCA Document Writing Board to ensure the initial intent as stated during the ORCA VTC held in Mar 07 was maintained as the primary objective. He reiterated that MOG 4 was delegated as the implementation authority for ORCA as directed by CMS, also in Mar 07.

2. He appealed to the group to ensure a balance was drawn between training and a qualification requirement that is safe vs. making the process too exclusive.

3. He also insisted that whatever recommendations were produced by this group that they flowed through the coastal authorities for oversight prior to be sent up within the chain of command at CMS.

4. As part of the discussions the topic of risk assessment was discussed and DMTE 2-2 contended that the responsibility of accepting the initial risk rests with CMS and not the coasts. Until the Navy resolves the OSS, QSP, qualification processes etc, that the responsibility of risk must be retained by the Centre.

5. Those that had not spent any significant time onboard an ORCA PCT spent the next 60-90 minutes touring and familiarizing themselves with the layout of ORCA with qualified SME providing insight into capabilities and limitations of the various ship's systems.

6. What did occur during this brief yet crucial period was the discussions in the margins with respect to manning requirements for various evolutions and emergencies.

7. MOG 4 staff provided an ORCA brief, which essentially set the stage of how we arrived at where we are with respect to ORCA capabilities, qualifications and training.

8. After lunch we began the review of the three OSS. Beginning with the OIC OSS, there was discussion as to why Boatswain should be an applicable MOSID for the OIC specialty and it was justified by the soon-to-be proposed requirement to support Navigation training for the Boatswain QL6A.

9. There was some discussion with respect to who should hold the responsibility of rescinding this qualification. The Chair's (DMTE 2-2) perspective was to treat it like any qualification and retain the control of rescinding it at the Centre however the coastal perspective was to draw a comparison with BWK and retain that responsibility at the operational authority level, i.e. CMOG 4

10. The remainder of the OIC OSS was amended to reflect the standard format for OSS. One new Task was added; Conduct Seamanship Evolutions. The remainder of the OSS was amended to group the Knowledge list to support those Tasks.

11. The 'Boatswain' OSS was renamed Senior Boatswains Mate. MARS, Marine Engineer and Clearance Diver occupations were removed from the list of applicable MOSIDs. Three prerequisites were removed; the requirement for 1st Aid, Cert 2 Marine Engineer and selection for employment on an ORCA. Two tasks were added; Operate Sea Boats and Respond to Emergencies. Two other tasks were reworded to reflect the recently acquired knowledge gained by those having received ICT.

Tuesday June 5, 2007

12. The forenoon of day two was used to review and amend the Marine Engineer OSS. The OSS submission by CFFSE contained 68 Tasks, which were reduced to nine. In reviewing the associated knowledge, there was some discussion in the amount of "theory" required to support a specialty for sailors who, arguably, should already possess the necessary theory to conduct the job-related tasks.

13. As expected, in reviewing the Marine Engineer OSS, there was substantial discussion regarding the baseline prerequisite of Cert 2 Marine Engineer vs. MESO QL3 Phase II (As would be the case for much of the subsequent 2 days of deliberations).

14. Being satisfied with the state of the three OSS, the afternoon of day two began with reviewing the first of the three QSP; the OIC. In reviewing and advising the group as to the traditional QSP writing process, the Chair had intended to try and follow that process as best as possible but it quickly became clear that the limited available time would not allow that process to work. Moreover, given that there were only 3.5 days to review three QSP, the decision was made to concentrate all efforts on ratifying only chapters one and two of each QSP. Therefore consensus amongst the group was to review chapters one and two of the three draft QSP, paragraph by paragraph, and amend as necessary to align with the correspondingly amended OSS, as opposed to conducting full Task and Instructional Analyses as defined by CFITES.

15. Discussion occurred to confirm the MITE Competency Code requirements from cradle-to-grave ORCA OIC training. On completion of the Tender OIC Course, for those occupations that require it, a MITE Code will be awarded. On completion of the ORCA class package (both academic and OJTPR phases), another MITE code would be awarded. Once one or both of these codes have been awarded – depending on the particular training requirement – the graduate would then be eligible to challenge the Tender OIC Board and conduct the Practical Check, both being conducted by MOG 4. Successful completion of both of these is the requirement for the OIC OSS National Qualification Code. As of the date of this ROP, none of these codes were assigned.

16. The board discussed the means by which the academic phase of the OIC training could be provided. The utility of Distance Learning was debated as well as whether or not a self-study package constituted a form of DL. The board conceded that while DL is always a possibility, it was too early to adequately identify components of the curricula, which could be delivered by DL. At the time of the ORCA document writing board, one particular candidate was being allowed the opportunity to study the ORCA ICT training material in order to bypass the academic phase of the class package. Discussion also occurred as to whether or not a time limit should be applied to any such self-study package and agreement was made that it was unnecessary.

17. On completion of the day, the Chair provided each board member with a copy of each of the revised OSS with direction to review during the evening and provide comment the following morning.

Wednesday, 6 June, 2007

18. The first order of business was to re-review each of the OSS, having individually reviewed them the previous evening. There were necessary changes to each of the OSS but they were all relatively minor in nature. The suggestion was made to provide the Marine Engineer with Small Boat Coxn training in order to provide redundancy should the SBM not be available. After some discussion, the decision was made to not allow the limited crew size to create scope creep with respect to cross-training requirements.

19. There was also the proposal to add “First Line Maintenance” as a Task, or at the very least as a Knowledge statement, to the Marine Engineer OSS. It was decided to include this level of maintenance as a teach point to each of the machinery operating Performance Objectives within the QSP. This debate also concluded to reduce the proficiency level of “Maintenance Documents and Procedures” from three to two.

20. It was identified that the reference authority to rescind the Marine Engineer qualification is captured in MARCORDS 9-17 (MESO) and 9-22 (Regular Force Marine Engineer).

21. Having been satisfied with the details of the three OSS, the board returned to the OIC QSP to complete its review. There was some debate regarding the course prerequisites, particularly how it would apply to CIC officers. It was understood that PO 004 of the TOIC Course would mirror the details of the OIC Class Package QSP thereby removing the requirement to conduct the class package training. A further note was added to the QSP to address those occupations that were already in possession of a Tender Charge Courses qualification.

22. Minimum manning of one was justified to allow the necessary flexibility to run the ORCA class package to limited incrementally tasked personnel.

23. The remainder of Chapter One of the OIC QSP was ratified without significant debate. Chapter Two, consisting of only one Performance Objective, was revised to better reflect the OSS.

24. A meeting occurred at the end of the forenoon, attended by:

- LCdr Bursey – DMTE 2-2;
- Cdr Boisjoli – DMPOR;
- LCdr Thomson – D Mar Pers;
- LCdr Van der Krogt – CFFSE MS Div Cdr;
- Lt(N) Johnston – MOG 4
- Lt Taylor – CFFSE TDO
- CPO1 Ewing – Marine Engineer Occupation Manager;
- CPO2 McLeod – CFFSE Standards
- CPO2 Pelletier – MARPAC F44
- PO1 Lachance – NAVRESHQ MESO
- PO2 Tremblay – MOG 4
- MS Ryan – MOG 4 ORCA Marine Engineer

The discussion revolved around whether or not MESO QL3 Phase II was sufficient enough to set the benchmark for ORCA Marine Engineer training. It was DGMPR staff assessment that the June 2006 letter from DGMPR (particularly paragraph 2), which stated that the MOG 4 establishment would be amended to Cert 2B to permit the applicable SPQR to reflect an ORCA endorsement, may have been misinterpreted. Since the letter did not include MESO, an assumption was made by CFFSE that the

training delta was not insignificant for a MESO to receive an ORCA Class Marine Engineer endorsement and as such the initial training package was designed solely for the Regular Force Cert 2. After substantial debate and subsequent direction from DMTE, the decision was made from the representatives of DGMPR that this board's work would be based on the OSS details; a valid Cert 2 certificate or MESO QL3 Phase II. Both MOG 4 and MARP F44 staff representatives supported this. A suggestion was made to CFFSE that if they had reservations that they should document and submit them up their Chain of Command. Consensus by everyone was that even with this disagreement, it would not preclude the board from reviewing the proposed ORCA Marine Engineer QSP.

25. The afternoon of day three was taken up by reviewing the ORCA Marine Engineer QSP. The Aim was rewritten to include the MESO occupation.

26. While it was agreed that timings could not be assigned until Chapter Four was completed, CFFSE requested guidance from within which to design their training. The Chair submitted that considering that MARPAC F44 agreed that given the course prerequisites and the recent success of ICT, that two weeks was not an unreasonable expectation, notwithstanding reservations with respect to the timelines from CFFSE. That said, it was also agreed that the OJTPR still required revision and as such the associated timeline could not be determined.

27. All the attending MARENG and MESO personnel attested to the requirement of a Cert 4 as a necessary staff resource to endorse completion of the Emergency Drills component of the OJTPR. Currently, there is no establishment billets for either a MSEO or Cert 4 at MOG 4 therefore resources are being drawn from either MARPAC or CCFP to achieve this certification requirement. Evidently, there are sufficient personnel resources available within the fleet to meet this requirement. This would not contradict the requirements within MARCORD 9-22.

28. CFFSE reported an opportunity to purchase an IPCS windows-based CBT license from L3 for $5K for one year. Notwithstanding that the proper process would need to be followed to purchase any such licenses for long-term CBT, it was agreed to be a valid requirement for future training. CFFSE was encouraged to submit the necessary paperwork to initiate the purchase process for these licenses.

29. Similar to the OIC QSP, the minimum loading was reduced to one student. However the maximum course loading for the Marine Engineer QSP was set at 10 students based on the necessity to share the same hull asset for the respective OJTPR requirements of the OIC and Boatswain QSPs; to maintain a respectable student-instructor ratio; to allow a reasonable time allotment to conduct the OJTPR; and because of the limited deck space within the ORCA engineering spaces.

30. The remainder of Chapter One of the ORCA Marine Engineer QSP was ratified without significant debate. The Chapter Two review commenced with a review of the draft scalar. Considering the multiple changes to the OSS, it was conceded that much of the scalar was OBE. To start, the Performance Objectives were renamed to reflect the changes to the OSS. After relatively minor amendments to PO 001 and 002, the day was adjourned.

Thursday, 7 June 2007

The board members were given the opportunity to review the ROP to date. Necessary amendments are already made above.

31. The board resumed work on the Marine Engineer QSP. There was some debate over the level of maintenance reports expected of the Marine Engineer. It was agreed that the MOG NTS would complete any formal engineering reports and returns.

32. Further discussion occurred regarding any presupposed timeline to deliver the Marine Engineer Class package. Figures had varied throughout the week from as little as two weeks to mirror the ICT to as long as six weeks (including OJTPR). Some members agreed that the Instructional Analysis need to be conducted to verify the true timeline required however, even the members of MARPAC F44 and MOG 4 NTS agreed that the two-week ICT was sufficient to endorse a Regular Force Cert 2 and MESO QL3 Part II Phase I with fulfilling the duties of an ORCA Marine Engineer.

33. CFFSE reminded the board of the recent CANFORGEN (?), which stipulated the Physical Training requirement for CF members. As a result, CFFSE issued a COTM that mandates three periods per week be injected into the current weekly training schedules.

34. Having completed the Marine Engineer QSP, the board started the Boatswain QSP review. The first order of business was to amend the required resources to mirror those of the other two QSPs.

35. The maximum course loading was mostly based on the available space in the Emergency Steering Gear compartment.

36. As discussed in the margins, the ROP will be submitted under DMTE signature, which will include direction for the Training Establishments (TE) to resubmit completed QSPs for approval under the respective Commandants' signature.

37. After reviewing the OIC QSP, the decision was made to delete any reference to DL until the module had been developed. Suffice to say, some components of the OIC training, for that matter all three positions, can be delivered via DL.

38. MOG 4 raised the question regarding minimum training requirement for the two Deck Hands. The initial proposal of NETP/NETPO may prove problematic when there is an anecdotal indication that STS(P) has imposed a regulation to have four personnel onboard with SCBA training. MOG 4 submitted that this is addressed by always deploying a 2 I/C to augment the five core crew. The 2 I/C requirement initiated further debate because it contradicted the COO. Consensus ended with a minimum requirement of NETP/NETPO minimum to fulfill the duties of Deck Hand, both of which *should* have the SCBA qualification to meet the 4 x SCBA qualification requirement of Sea Training Staff.

39. Discussion occurred as to a potential requirement to 'certify' the OIC to react to engineering emergencies using the BCC/IPCS. This will require a Cert 4 to accompany the OIC during his/her OJTPR to assess the engineering drills.

40. Secretarial discussion occurred to address the prerequisite of CIC, Boatswain QL6A/Reserve QL3 and Clearance Diver QL6B to receive the ORCA Class Package as part of the Tender OIC (TOIC) Course. It became impossible to come to any common solution regarding this recommendation considering that there was no firm plan to replace the Diving Tenders with ORCA hulls #7 and #8, i.e. MOG 5 Clearance Divers would be forced to conduct part of their OIC training at NOTC even when employed in SEA SCHELDT. That said, in the fullness of time, this will likely result in NOTC VENTURE accounting for an additional production requirement into their annual Capability Plan, i.e. FDU(A) will eventually need to qualify future Tender OIC onboard ORCA.

Friday, 8 June 2007

41. Having been given copies of the three QSP and OSS the previous night, each board member was provided an opportunity to suggest revisions / corrections. The most significant change was the addition of the subparagraphs to the OIC Standard statement. Although having only one Performance Objective was deemed satisfactory, elaboration was required in order to provide sufficient details to assist training design and better define was meant to be observable and measurable within the Standard statement.

42. After the board members completed the changes to each of the six documents, those not involved with the Marine Engineering training were thanked for their exceptional contribution throughout the week and dismissed for the weekend. The remainder of the forenoon was focussed on reviewing MARCORD 9-17 and 9-22. The bulk of the initial discussion revolved around the necessity for any type of ORCA 'endorsement' on an existing Certification when it was the position of the board Chair that the National Qualification Code for the ORCA Marine Engineer should be all that was necessary. For Regular Force MARENG it is less complicated because we've set the baseline at a specific certificate level – Cert 2. For MESO the baseline is set *between* Tickets so the question was raised as to which Ticket should an ORCA *endorsement* be assigned. Consensus was to create a separate ORCA MESO Ticket altogether and add that to paragraph four of MARCORD 9-17.

43. The remaining board members recommended a separate review be conducted to align MARCORD 9-17 and 9-22 with the proposed ORCA Marine Engineer qualification and training requirements.

44. In addition to the above task, the board recommended that this ROP be delivered under cover letter from DMTE with further instructions for the two CCA (CFFSE and NOTC) to complete Chapters 3 and 4 and to concurrently investigate the feasibility of DL when conducting the necessary training methodology analyses.

Monday, 11 June 2007

45. The first order of business was to review MARCORD 9-50, Annex E. After some philosophical discussions with respect to how MOG 4 is currently endorsing an OIC to assume responsibility for ORCA versus how that practice should be reflected in 9-50, the board members decided to read through the latest draft of 9-50 Annex E (Tender OIC Qualifications) first.

46. The decision was made to delete any distinction between Tender Charge and Tender Command. For the purposes of this MARCORD, the responsibilities do not change. Any restrictions placed on what tasks, i.e. missions, can be accomplished by certain OIC would be captured under the Concept of Operations.

47. There was some discussion as to the justification for the 42 days at sea requirement for Clearance Diver, Boatswain and CIC. It was decided that until the OJTPR portion of the Tender OIC Pilot Course is complete, that there was insufficient SME available on the board to dispute the validity of 42 days. What's more is that it was unclear as to the Clearance Diver QL6B OJTPR requirements and how it compares to the Tender OIC QSP.

48. Discussion occurred regarding the SRI III process for the core ORCA crewmembers. The MOG 4 representative briefed that each individual has to pass an SRI III but there is no requirement to conduct a Team SRI III. After discussion with Cdr Leblanc from STS(P), there was some debate as to this requirement. In the end, it was still unclear as to the STS requirement to conduct even a rudimentary 2-4 hour SRI III 'check' to verify the skill sets of the 3-5 crew members to safely operate a Tender as a cohesive team.

49. Much discussion occurred regarding the previous week's recommendation to inject the ORCA Class package into the Tender OIC Course. The argument against this requirement applies to all classes of Tender other than ORCA and to the utility of forcing, for example, a Boatswain that is about to be employed in the Nanoose Range vessel with having to conduct an ORCA OJTPR or a Clearance Diver employed in YDT Sooke or Secheldt. It was decided that the Tender OIC QSP would need to be modularized such that not everyone who is eligible to take the training would need to take the ORCA module. This would further require a separate OSS for the Diver Tenders and one for all other Tenders. This prompted the board to add the comment, "The Tender OIC Qualification is valid only for the particular class of tender for which the qualification was originally awarded", into 9-50. Further discussion concluded that 9-50 Annex E did not apply to the Range vessels as these vessels met the definition of "Small Craft". The question remaining was what training and qualification process was/is in place to address this size of vessel. A further complicating factor was the multitude of small craft and tenders listed in MARCORD 10-02. Does the Naval Training System need a separate training and qualification process for each of these as well?

50. Deleted any reference in 9-50 regarding the use of ORCA to conduct MARS IV. It was decided that this would best be captured in both MARCORD 4-26 and/or the COO.

51. The Training Officer from FDU(P), Lt(N) Vidalis, was given the opportunity to review the recommended changes to 9-50 to comment on how they impact Clearance Diver training. It was verified that the QL6B QSP already contains the curricula to meet the Tender OIC requirement. I asked FDU(P) to liaise with NOTC and compare the Clearance Diver QL6B and TOIC QSPs. It was secretarially discovered that, in fact, it well exceeds the TOIC curricula. It is foreseen that either the Tender OIC component of the Clearance Diver QL6B will be extracted and those students will conduct the TOIC Course or the TOIC QSP will need to be amended to reflect what is in the Clearance Diver QL6B QSP, pending the results of the TOIC Pilot Course validation. This may translate into NOTC, as the TOIC CCA, having to add this requirement to their CP. The

additional funding requirement, albeit still undetermined but guessed to be not more than $30K, was estimated to have a negligible overall impact on NOTC.

Tuesday, 12 June 2007

52. Cdr Leblanc from STS(P) joined to assist the board in reviewing MARCORD 45-06. After some discussion regarding the necessity regarding SRI III, the board agreed that DMTE should instruct the CCAs, i.e. CFFSE and NOTC to ensure their respective QSP PCs, match the SRI III requirement.

53. The Concept of Operations consumed the remainder of day two. Significant rewriting occurred but the most contentious issue was the various interpretations of the minimum crewing requirements to achieve the different potential missions; training or otherwise. Clearly there was a difference of opinion and arguably a miscommunication between the Operational Authority and the Centre regarding how many crewmembers will be required to achieve anything other than rudimentary deployments.

54. Several questions were raised which will require another review of the TOIC Course. Should the CIC be required to complete NETPO in total or only the DC Module? It was argued that the CIC already receive other training, which potentially encompasses the curricula taught on the NETPO Course. NOTC will be conducting a comparison to identify any possible delta, which will need to be inserted in the CIC training.

55. When discussing the various crewing requirements, it became very clear that different agencies were interpreting the various requirements differently. There is an acceptance that five is the minimum number of people required to safely take an ORCA to sea. The question pertained to the tasks that this crew could actually achieve. As soon as anything from a benign training mission up to an operational mission are assigned, the crew requirements increase exponentially. As a result, a table was designed to capture the crew requirements based on the time away from port. This table was inserted into the Concept of Operations instead of writing out the multiple levels of missions versus crew matrix.

Wednesday, 13 June 2007

56. The board continued with the review of the Concept of Operations. The most significant change from the previous day was the introduction of the Crew Matrix Table. This received support from those in attendance, which included representation from NOTC and MOG 4.

57. A final review was conducted of the whole COO with minor changes being made but overall the document was completed with general consensus of the board.

58. The board had an opportunity to brief CMOG 4 with the results of the OSSs, QSPs, MARCORDs and COO to date. He expressed serious concern regarding the achievability of the proposed minimum crew requirements written in the COO Table, particularly the requirement for 13 crewmembers when the vessel is operating in excess of 16 hours. This number was conceived based on a 1-in-3 watch rotation and did not account for anchor watches. Arguably the minimum crew size could be scaled back somewhat if the vessel is only operating 8-12 hours per day and going to anchor each

night; another row within the table may inevitably be required. In the end, Capt(N) Truelove guaranteed the board that MOG 4 feedback would be provided.

59. Select members of the board met with Cmdt CFFSE and his staff, namely those of the MSE Division; to discussion reservations they have with the condensed process being followed to conduct a proper instructional analysis in support of the Marine Engineer QSP. He expressed worry that by rushing the process in answer to various external pressures that key information would not be captured and an untenable training product would be produced. What's more, he conceded that CFFSE does not have the requisite MESO SME to adequately identify any potential training delta for the MESO QL3. As such, it is intent to continue work towards delivering a QSP to solely address training a MARENG Cert 2. This QSP is expected to be ready for approval during the summer months following this board and, with the necessary MESO support from NAVRES, a training delta product (again, assuming one is even required) for MESO QL3.

Thursday, 14 June 2007

60. Transport Canada's TP2293E was reviewed to determine the delta between an ORCA OIC and the TC requirements for a Master Mariner 500gt Local Voyage. Unfortunately, during the board the TC website had removed all references to the Nautical Certificates requirements nor had they updated their website to reflect Master 500gt. As such, the board was forced to use the BCIT site, as it's primary resource. BCIT, being a recognized maritime training institution for TC, was deemed to be an acceptable source of information for the board to identify the majority – if not all – of the delta. The results show that the Navy covers the majority of the academic requirements but do not meet the Sea Service requirements of TC.

61. The board concluded all business by lunch and adjourned.

ORCA PILOT COURSE

Dates: 1 – 15 Oct 2007

Location: Esquimalt BC

Course attendees:

Instructor – Lt (N) Jeff Chura

Class - LCdr Crawshaw, Lt (N) McFarlane, Lt(N) Harwood, Lt(N) Darville, & Lt(N) Sutherland.

Training:

First week was classroom work which was held on board an ORCA class vessel.

OBSERVATIONS	ACTION
Sequence of instruction did not flow to maximize the effectiveness of classroom	Realigned schedule.

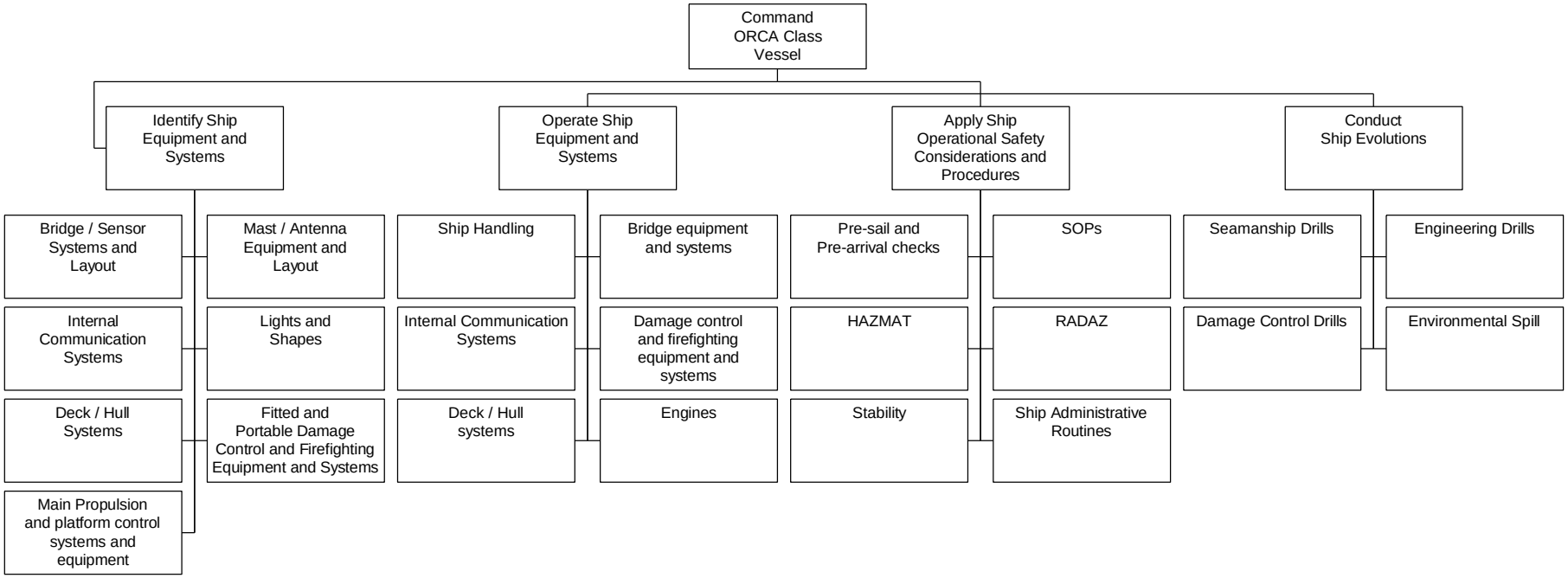
training.	
PIP was needed for the characteristics of the ship and should be handed out prior to the start of the course.	Too be promulgated.
A walk through of the emergencies is needed as they are being taught.	Incorporated into TP.
There was information given in the classroom that was not needed for the OIC course, for example the valves lecture.	Incorporated.
Information like redundancies and liquid levels are important to have in the lectures.	Incorporated.

SECOND WEEK OF OIC TRAINING WAS AT SEA.

OBSERVATIONS	**ACTION**
The students felt that the coming alongside practicing was important to their learning the characteristics of the ORCA vessel.	Status Quo
Each student being OIC for the day was a great teaching method.	Status Quo
There was a need to incorporate into the training where the OIC runs the emergencies drills with minimum manning.	Incorporated into the training schedule. The drills to include anchorages, buoy approaches, MOB and engineering drills.

ANNEX D – QUALIFICATION STANDARD AND PLAN WRITING BOARD SCALAR

ORCA OIC Class Package Scalar



Approval Date: 06 Mar 15
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ANNEX E – OJTPR

ORCA OIC CLASS PACKAGE

PRACTICAL TRAINING REQUIREMENTS (PTR)

FOREWORD

62. This Practical Training Requirement (PTR) package is designed to provide details for the employment of trainees during the ORCA Phase II.

63. The publication is UNCLASSIFIED and is issued under the authority of the Commanding Officer, VENTURE, The Naval Officers Training Centre. It is effective immediately and supersedes all previous versions.

64. Recommendations for amendments are to be forwarded to the Commanding Officer, VENTURE NOTC through the Standards Department.

RECORD OF AMENDMENTS

REVISION	APPROVAL DATE	AUTHORITY

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PRACTICAL TRAINING REQUIREMENTS FOR ORCA CLASS AFLOAT TRAINING PHASE

AIM OF THE AFLOAT TRAINING PHASE

65. The aim of the ORCA Class Afloat Training Phase is to introduce the trainees to the unique characteristics of this class of ship. The trainee is shown how the ship handles, equipment onboard and drills expected of the OIC of an ORCA Class.

PRACTICAL TRAINING REQUIREMENTS (PTR) AND ENABLING OBJECTIVES (EOS)

66. The PTRs for this training are contained in Part Two of this chapter, numbered sequentially by topic. Details of the EOs can be found in Chapter Four of the ORCA OIC QSP.

67. The PTRs listed in Part Two are to be completed as pre-requisites for successful completion of training. Instructions for assessment of the trainee completing the PTRs are contained in Chapter Three of the ORCA OIC Class Package QSP.

PTR INSTRUCTIONS

68. As the trainee completes each PTR to the required standard, the designated assessing Officer shall sign and date the applicable item in PTR Tracking Section.

69. Should the Trainee be unable to complete any part of a PTR due to the training limitations of the platform, then he/she shall complete the requirement either verbally or by completion of an assignment/project assigned by the CTO. Once the individual has demonstrated a sound understanding of the material, then the requirement may be signed off.

70. When the PTR shows that an individual has completed all of the required performance objectives, the Completion of the Requirements signature block is to be signed by the Course Training Officer responsible for the training.

PTR SIGNING AUTHORITIES

71. The CTO and/or CO will normally sign the PTRs. The following table indicates the specific signing authorities. VENTURE may authorize additional authorities depending on the staff carried onboard. (Note that when XO is indicated, the XO must be ORCA OIC Certified).

PTR #	SIGNING AUTHORITY
BRIDGE EQUIPMENT AND SYSTEMS	
001-014	CTO/CO
SEAMANSHIP	
020- 027	CTO/CO
ENGINEERING	
030-031	CTO/CO/SI
DAMAGE CONTROL	
040-041	CTO/CO/SI

PTR COMPLETION TRACKING

72. The Practical Training Requirements (PTRs) listed herein are to be completed as pre-requisites for successful completion of the ORCA OIC Class Package Part II - OJTPR.

73. This section contains the following PTRs:

- Bridge Equipment and Systems 001 – 019
- Seamanship Drills 020 – 029
- Engineering Drills 030 – 039
- Damage Control Drills 040 – 049

PTR PROGRESS SHEETS

74. The following PTR Progress Sheets are to be filled out and signed by the appropriate authority to assist the trainees and signing authorities to keep track of numbers of watches/tasks completed towards PTRs:

BRIDGE EQUIPMENT AND SYSTEMS PTRS

PTR #	PERFORMANCE	1	2	3	DATE	INITIALS
001	Radar systems					
002	Radio systems					
003	ECPIN					
004	GPS					
005	Autopilot					
006	AIS					
007	IPCS					
008	Echosounder					
009	Speed Log					
010	Meteorological System					
011	Light Panel					
012	Gyro repeats					
013	Magnetic compass					
014	Ship whistler					

SEAMANSHIP

PTR #	PERFORMANCE	1	2	3	DATE	INITIALS
020	Launch and recover RHIB					
021	Anchoring, mooring and coming to a bouy					
022	Departing					
023	Berthing					
024	Man Overboard					
025	Towing					
026	Blind Pilotage					

027	SAR					

ENGINEERING

PTR #	PERFORMANCE	1	2	3	DATE	INITIALS
030	Steering gear breakdown					
031	Engineering emergencies, alarms and difficulties					

DAMAGE CONTROL

PTR #	PERFORMANCE	1	2	3	DATE	INITIALS
040	Fire					
041	Flood					

COMPLETION OF THE REQUIREMENTS

The Officer under training has completed all the requirements of the Training Standards for the ORCA OIC Class Package Part II – On the Job Training Practical Requirements (OJTPR).

(Rank)	(Name - Print)	SN
(Date)	(CTO Name & Signature)	(Rank)

PTR REQUIREMENTS

BRIDGE EQUIPMENT AND SYSTEMS

75. Performance. Operate bridge equipment and systems.

76. Conditions:

a. given: ORCA Class ship.

b. denied:

(1) references; and

(2) assistance;

c. environmental: At sea or alongside in any or all environmental conditions.

77. Standards. The candidate shall safely and correctly operate the ORCA Class vessel's bridge equipment in accordance with SSOs to include:

(1) Radar systems;

(2) Radio systems;

(3) ECPIN;

(4) GPS;

(5) Autopilot;

(6) AIS;

(7) IPCS;

(8) Echosounder;

(9) Speed Log;

(10) Meteorological System;

(11) Light Panel;

(12) Gyro repeats;

(13) Magnetic compass; and

(14) Ship whistler.

SEAMANSHIP DRILLS

78. Performance. Demonstrate seamanship drills

79. Conditions:

a. given: ORCA Class ship

b. denied:

(1) references; and

(2) assistance

c. environmental: At sea or alongside in any or all environmental conditions.

80. Standards. The Officer shall, safely and in accordance with the SSOs and practice of good seamanship, perform seamanship drills:

a. Launch and recover RHIB;

b. Anchoring, mooring and coming to a bouy;

c. Berthing and Departing;

d. Man Overboard;

e. Towing;

f. Blind Pilotage; and

g. SAR.

ENGINEERING DRILLS

81. Performance. Demonstrate engineering drills.

82. Conditions:

a. given: ORCA Class ship;

b. denied:

(1) references; and

(2) assistance;

c. environmental: At sea or alongside in any or all environmental conditions.

83. Standards. The Office shall safely perform engineering drills in accordance with SSOs to include,

a. Steering gear breakdown; and

b. Engineering emergencies, alarms and difficulties;

DAMAGE CONTROL DRILLS

84. Performance: Demonstrate damage control drills.

85. Conditions:

a. given: ORCA Class ship;

b. denied:

(1) references; and

(2) assistance;

c. environmental: At sea or alongside in any or all environmental conditions.

86. Standards : The candidate shall safely in accordance with SSOs

a. Fire; and

b. Flood

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ANNEX F – ACRONYMS / ABBREVIATIONS

ACRONYM / ABBREVIATION	TERM/DEFINITION
AIS	Automatic Identification System
ARPA	Automatic Radar Plotting Aid
AX	Quarterdeck of the ship
BWK	Bridge Watch Keeping
BR45	Admiralty Manual of Seamanship
C/S	Call Sign
CCA	Curriculum Control Authority
CIC	Cadet Instructor Cadre
CMS	Chief of Maritime Staff
CONOPS	Concept of Operations
CP	Cable Party
DC	Damage Control
ECPINS	Electronic Chart Precise Integrated Navigation System
ECDIS	Electronic Chart Display Information System
EO	Enabling Objective
E/S	Echo Sounder
GMDSS	Global Maritime Distress and Safety System
GPS	Global Positioning System
HAZMAT	Hazardous Material
HF	High Frequency
HVAC	Heating Ventilation Air Conditioning
IAW	In Accordance With
IRB	Inflated Rescue Boat
INFP	Immediate Notice For Power
IPCS	Integrated Platform Control System
MA	Managing Authority
MCTS	Maritime Communication and Traffic System
MOB	Man Overboard
MOC	Maritime Operations Centre
MOG	Maritime Operations Group

ACRONYM / ABBREVIATION	TERM/DEFINITION
NFU	Non-Follow Up
NTS	Naval Tender Section
NOTC	Naval Officers Training Centre
OIC	Officer In Charge
OJTPR	On Job Training Progression Report
OS	Occupation Specification
OSS	Occupation Specialty Specification
PO	Performance Objective
PSH	Part Ship Hands
PTR	Progression Training Requirements
QSP	Qualification Standard Plan
RADHAZ	Radiation Hazard
RFLG	Ready For Letting Go
RHIB	Rigid Hull Inflated Boat
ROTR	Rules Of The Road
SAR	Search and Rescue
SCBA	Self Contained Breathing Apparatus
SOP	Standard Operating Procedures
SSD	Special Sea Duty
TOIC	Tender Officer In Charge
TRB	Training Review Board
UPS	Uninterrupted Power Supply
VHF	Very High Frequency
VTC	Video Teleconference

ANNEX G – SUGGESTED SEQUENCING AND TIMINGS

(in 50 min. periods)

PO / EO	OBJECTIVE	Class	Sim	Other	Sea Phase		EC	PC	PO
					Dev	Eval			Total Periods
PO 001	COMMAND ORCA CLASS VESSEL								
EO 001.01	Identify Ship Equipment and Systems	11				0			11
EO 001.02	Operate Ship Equipment and Systems	14				0			14
EO 001.03	Apply Ship Operational Safety Consideration and Procedures	9				0			10
EO 001.04	Conduct Ship Evolutions				14	21			35
OJPR									
OIC Certification									
	TOTAL PERIODS								
	TAT (in periods)	0	0	0	0	0	0	0	0
	TOTAL COURSE TIME (in weeks)								2

Time Table Week 1

ORCA Class OIC Course (Afloat Phase)					
	Monday	Tuesday	Wednesday	Thursday	Friday
8:00	Introduction OIC Instructor	DC equipment	Fuel Systems	Ship's Service Air	SOPs Part Three-Emergencies
8:15					
8:30	Break				
8:45	IPCS			Internal Communications	Break
9:00					Practical FF/DC walk through: incident boards, attack routes, hoses, etc.
9:15			Break		
9:30			Prop and Auxs	Break	
9:45	Break	Break		External Communications	
10:00	Ship's Layout	Bilge stripping			
10:15					
10:30					

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10:45		Break		Break	
11:00		HVAC		Navigation Equipment	
11:15	PT Period		PT Period		PT Period
11:30					
11:45					
12:00	Lunch 1200-1300				
13:00					
13:15	Hull fittings	Domestics	Sea Water Cooling	SOPs Part One—Ship's Organization	Knowship Books/OJPTRs/Preps for sea phase/Open questions
13:30					
13:45					
14:00				Break	
14:15	Break		Break	SOPs Part Two-Operations	
14:30	Deck Work		Power Generation & Distribution		
14:45					

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15:00				Break	
15:15		Break			
15:30	KY Ship Books	Steering		SOPs Part Two (Cont.)- General Safety:	
15:45					
16:00					

Time table Week 2

	Monday	Tuesday	Wednesday	Thursday	Friday
8:00					
8:15					
8:30					
8:45					
9:00					
9:15					
9:30					
9:45	EO 001.4	EO 001.4	EO 001.4	EO 001.4	EO 001.4
10:00	Sea Developemental	Sea Developemntal	Sea Developement	Sea Assessment	Sea Assessment
10:15	Use check list in Annex K	Use check list in Annex K	Use check list in Annex K	Use check list in Annex K	Use check list in Annex K
10:30					
10:45					
11:00					
11:15					
11:30					
11:45					

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12:00-1300	LUNCH				
13:15 13:30 13:45 14:00 14:15 14:30 14:45 15:00 15:15 15:30 15:45 16:00	EO 001.4 Sea Developemental Use check list in Annex K	EO 001.4 Sea Assessment Use check list in Annex K	EO 001.4 Sea Assessment Use check list in Annex K	EO 001.4 Sea Assessment Use check list in Annex K	EO 001.4 Sea Assessment Use check list in Annex K

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ANNEX H – TRAINING RESOURCES

In accordance with Chapter 1 paragraph 3 add any extra tables or paragraphs used for Resources in this annex. (**Not** to be deleted.)

ANNEX I – COSTING INFORMATION

Include information about cost in whichever format is applicable.

This annex is optional (**Not** to be deleted).

ANNEX J – COURSE HISTORY

87. This annex is to record course history issues such as explanations of amendments, pertinent findings, trends, resource problems and modifications. Focus should be given to both excellent and questionable issues of the course. Capturing important course history information is a useful managerial tool for amendment considerations by future course developers.

88. **Note: This annex shall NOT be deleted**. This annex must be annotated with every update / modification to this QSP. This is in addition to making the appropriate entry in the Record of Amendments table. Detail as to the nature and reasons for amendments is to be provided here.

Amendment 06 Mar 2015

Amendment to QSP required due to wrong NQual being listed in the forward. Secretarial changes made to align with NPTORD. Ch 1 & 2 authorities changed to reflect proper authority.

ORCA_OIC_ApprovalSheet.pdf

ANNEX K – CHECKLISTS

ORCA At Sea Drills

- all exercises/items in OJPTRs*

- Departures

- Approaches: ahead/astern; port/stbd

- turning at rest both directions

- Coming to full speed ahead from stopped

- coming to full speed astern from stopped

- stopping from full speed by simply stopping telegraphs and drifting

- stopping from full speed by going full speed astern

- going from full speed astern from full speed ahead

- turning to port/stbd at various speed with both engines

- turning to port/stbd at various speeds by stopping in-board engine

- turning 360 degrees at various speeds to see tactical and final diameter

- bouy approaches/departures

* Half of the students should do with minimum manning and half with all available resources, so all are exposed to considerations for both types of crew deployments.

ORCA Class Officer in Command Practical Assessment

ITEM	OBSERVED	COMMENTS
PRE-SAIL CHECKS		
Promulgated comprehensive Night Orders		
Time for INFP		
Required calls/reports		
Special instructions		
PRE-SAIL SAFETY BRIEF		
Intentions		
Priority line		
Additional fenders (as required)		
Special instructions (e.g. “AX report any small contact as we back away…”)		
SSD WARNING & CLOSING UP PIPES		
Report from Engineer		
Immediate NFP		
Defects		
Fuel/water %		
Shore connections		
Report from Buffer		
Ready in all respects		
Anchor RFLG		
Sea boat ready for sea		
All personnel onboard		

ITEM	OBSERVED	COMMENTS
BRIDGE EQUIPMENT SETUP		
Gyros checked		
E/S, Log, radar & ECDIS operating		
Transfer control of main engines		
'TROLL' feature selected		
Telegraphs under positive control		
COMMUNICATIONS		
Appropriate VHF calls made		
C/S bent on & hoisted		
DEPARTURE		
Demonstrated firm control of bridge team		
Control of part of ships and line-handling		
Accurate ship-handling & precise conning		
Appropriate sound signals		
Complied with local regulations (speed & wake considerations)		
Contact Avoidance/ROTR		
Remained clear of navigational dangers		
Maintained Situational Awareness		
COASTAL NAVIGATION		
Adequately applied principles of BR45		
Complied with MCTS requirements		
Knowledge of NTS SOPs and application of regulations (e.g. weather restrictions)		
Established a fixing routine		
Maintained proper lookout		

ITEM	OBSERVED	COMMENTS
Correct knowledge/application of PCT conning orders		
Appropriate use of Auto-pilot		
Maintained accurate records (OOW notebook & ship's log)		
Contact avoidance/ROTR		
Monitored weather and state of visibility		
Conducted Restricted Visibility check-offs		
REACTED TO EMERGENCIES		
Overall		
Established clear Command Aim and Command Priorities		
Ensured safety of personnel		
Maintained good situational awareness		
Maintained good information flow		
Always applied a solid 'Command Perspective'		
Steering Gear Failure		
Adequate knowledge of steering system (e.g. switched to NFU mode first)		
Remained in control of ship (steered by main engines/stopped ship)		
Appropriate pipes, sound signals, shapes & flags		
Correctly established control of rudders from aft steering		
Established reason for failure		
Transferred control of rudders back to the bridge		

ITEM	OBSERVED	COMMENTS
Secured aft steering		

Man Over Board		
Correct initial response (i.e. pipe, turn etc.)		
RADHAZ		
Accurate conning		
Maintain visual contact with MOB (control of lookouts)		
Communications with AX		
Safely launched boat (i.e. proper lee, reduced roll, steady co/spd etc.)		
Control of boat (e.g. Comms to vector boat to man/rx report of condition etc.)		
Safe recovery onboard & treatment of casualty		
CASEVAC considerations & reporting protocols		
Damage Control Exercise		
Correct initial response (*i.e.* pipe, CO_2 drench *etc.*)		
Correct preps for CO2 drench incl Command Approval		
Accounted for personnel/Mustered in a safe location		
Control of incident (*i.e.* established boundaries, info flow)		
Made accurate SITREP pipes		
Maintained an incident board		
Ensured ventilation & power crash stopped		
Safely manouevred ship		

Relative wind		
Dealt with casualties		
Shut down engines		
Shapes		
Anchor		
Abandon ship		
PILOTAGE, ANCHORING AND ENTERING HARBOUR		
Pilotage		
Basic planning safety (noting that there is no officer borne for navigation duties)		
HOT calculated & echo sounder monitored		
LDL and Clearing bearings on chart		
Use of transits and bearings to determine ship safety/position		
SSD closed up or anchor RFLG		
Conning and ship-handling		
Anchorage		
Basic planning safety (noting that there is no officer borne for navigation duties)		
HOT calculated & echo sounder monitored		
LDL and Clearing bearings on chart		
Use of transits and bearings to determine ship safety/position		
SSD closed up or anchor RFLG		
Conning and ship-handling		
Entering Harbour		

Demonstrated firm control of bridge team		
Control of part of ships and line-handling		
Accurate ship-handling & precise conning		
Appropriate sound signals		
Complied with local regulations (speed & wake considerations)		
Contact Avoidance/ROTR		
Remained clear of navigational dangers		
Maintained Situational Awareness		
COMMAND APPRECIATION		
General ship knowledge (max cap, location of main lifesaving equipment, cap of life-rafts, min crew size, technical parameters *etc.*)		
Situational Awareness		
Prioritization		
Communication		
Safety of personnel		
Safety of the ship		
Knowledge of own limitations		

	Pass/Fail
Overall Assessment:	
Comments:	

PTR #	PERFORMANCE	1	2	3	DATE	INITIALS
001	Radar systems					
002	Radio systems					
003	ECPIN					
004	GPS					
005	Autopilot					
006	AIS					
007	IPCS					
008	Echosounder					
009	Speed Log					
010	Meteorological System					
011	Light Panel					
012	Gyro repeats					
013	Magnetic compass					
014	Ship whistler					

SEAMANSHIP

PTR #	PERFORMANCE	1	2	3	DATE	INITIALS
020	Launch and recover RHIB					
021	Anchoring, mooring and coming to a bouy					
022	Departing					
023	Berthing					
024	Man Overboard					
025	Towing					
026	Blind Pilotage					
027	SAR					

ENGINEERING

PTR #	PERFORMANCE	1	2	3	DATE	INITIALS
030	Steering gear breakdown					
031	Engineering emergencies, alarms and difficulties					

DAMAGE CONTROL

PTR #	PERFORMANCE	1	2	3	DATE	INITIALS
040	Fire					
041	Flood					

COMPLETION OF THE REQUIREMENTS

The Officer under training has completed all the requirements of the Training Standards for the ORCA OIC Class Package Part II – On the Job Training Practical Requirements (OJTPR).

(Rank)	(Name - Print)	SN

(Date)	(CTO Name & Signature)	(Rank)